Mothukuri Greeshmika

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.94
Narayana Junior College	Class XII (SSC)	2024	98.40%
The Creek Planet School	Class X (CBSE)	2022	98.40%

SKILLS

Programming & Scripting Languages: C/C++, SQL, HTML, CSS, Javascript, Go

Tools, Libraries & Frameworks: Git/GitHub, PostgreSQL, ReactJS, Tailwind CSS, Gin, LaTeX

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures & Applications, Algorithms, Theory of Computation, Compilers, Database Management Systems.

Mathematics: Probability, Applied Statistics, Linear Algebra, Convex Optimization, Ordinary Differential Equations, Number Theory, Calculus I & II, Complex Variables, Series of Functions, Vector Calculus.

PROJECTS

Finding Nearest Neighbours using Voronoi Diagram | C++ **|** *DSA Lab Project* **|** GitHub Link

- Implemented **Fortune's Algorithm** in **C++**, utilizing a **priority queue** for site and circle events (**sweep line**) and a search tree to maintain the **beach line** during Voronoi diagram construction.
- Built an **adjacency graph** of Voronoi cells and developed an efficient traversal algorithm using **half-plane tests** to navigate neighboring regions and identify the nearest site for arbitrary query points.
- **Optimized search time** by restricting computations to locally adjacent Voronoi cells instead of exhaustively evaluating all input points, improving practical query **efficiency** over brute-force approaches.
- Validated the **correctness** of Voronoi cell construction and nearest-neighbour queries through extensive testing with multiple point distributions.

Statistical Hypothesis Testing | *Applied Statistics Course Project* **|** Project Report

- Performed **two-tailed, left-tailed, and right-tailed hypothesis tests** along with the **Chi-square test** of independence on real-world datasets to evaluate statistical claims and relationships between variables.
- Computed **test statistics**, **critical values**, **p-values**, and **rejection regions**, interpreting statistical significance at a **5% significance level** to draw data-driven conclusions.
- Applied **inferential statistical** techniques to analyze study hours, waiting times, battery life, and categorical datasets, validating hypotheses through statistical evidence and drawing meaningful data-driven conclusions.

Digital Time Capsule *(Ongoing)* **|** React, Go (Gin), PostgreSQL, Google OAuth **|** GitHub Link

- Developing a **full-stack** web application that enables users to securely create, organize, and manage **digital time capsules** through an intuitive and responsive user interface.
- Implemented **Google OAuth 2.0** authentication with **automatic user registration**, securely persisting authenticated user profiles in **PostgreSQL** .
- Implemented **scheduled capsule unlocking** and notification workflows, allowing users to receive access to their capsules only after a specified release date.

To-Do Application | *React, Go, PostgreSQL* **|** GitHub Link

- Developed a full-stack task management application using **React (Vite)**, **Go (net/http)**, and **PostgreSQL**, enabling users to efficiently organize daily tasks through an intuitive and responsive interface.
- Designed and **implemented RESTful CRUD APIs** to create, retrieve, update, and delete tasks with persistent

PostgreSQL storage, ensuring reliable data management across sessions.
- Built a **calendar-based interface** to organize and retrieve tasks by date, featuring **progress tracking**, **task completion status**, and **date-wise task visualization** for improved usability.
- Integrated the **React frontend** with the **Go backend** using **asynchronous API calls**, and managed application configuration through **environment variables** for flexible deployment across different development environments.

Music Store Database Analysis | SQL, PostgreSQL **|** *DBMS - I Course Assignment*

- Built analytical **SQL queries** to solve 10 real-world problems using **PostgreSQL**, extracting meaningful insights from a relational music store database.
- Utilized **CTEs**, **JOINs**, **GROUP BY**, **HAVING**, **UNION**, **INTERSECT**, **aggregate functions**, and **nested subqueries** to perform complex data analysis and reporting.

POSITIONS OF RESPONSIBILITY

Core Member, Lambda - *Software Development Club.*

- Contributed to the development of frontend for the **KRC Admin Console** using ReactJS.
- **Hosted a session** on ReactJS for students, introducing core concepts.
- Collaborated with team members to propose feature ideas for Alumni website.

Head of Operations, Abhyanth - *Annual Sports Fest of IITH*

EXTRACURRICULAR

Basketball: Represented **IITH** in the **Inter IIT Sports Meet 2024,2025** competing against teams from **23 IITs.**

Abacus: Completed level 10 and participated in multiple National Abacus Competitions.

Art: Participated in multiple art competitions, won **2nd prize** at the **Telangana Forest Academy** painting competition.

Taekwondo: Earned the **purple** belt (rank 5) in Taekwondo.